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# Robotic thyroid ultrasound scanning based on adaptive force/position hybrid control

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**Abstract.** To address the issues of unstable contact force and operator dependence in traditional thyroid ultrasound examinations, this paper proposes a robotic ultrasound scanning method based on adaptive force/position hybrid control. A 6-DOF robotic manipulator (xArm6) integrated with a six-dimensional force sensor is employed. Considering the anatomical structure of the thyroid, a "star-pattern" scanning trajectory is designed to ensure comprehensive coverage. An admittance control algorithm based on a second-order mass-spring-damper model is implemented to regulate the contact force, setting a constant target force of 5.0 N. Experimental results on a healthy volunteer demonstrate that the system exhibits a fast step response with a settling time of approximately 0.73 s. During continuous scanning, the contact force remains stable with a Root Mean Square Error (RMSE) of less than 0.25 N, significantly improving the stability and standardization of the examination.

## 1. Introduction

Thyroid nodules are a common clinical condition, and ultrasound is the primary modality for diagnosis. However, conventional manual scanning heavily relies on the radiologist's experience. Inconsistent contact force can lead to tissue deformation, affecting image quality and diagnostic accuracy [1]. Furthermore, the repetitive nature of the scanning task poses a risk of musculoskeletal disorders for sonographers.

In recent years, robotic ultrasound systems (RUS) have gained attention worldwide for their potential to standardize the scanning process. Research groups from different countries have proposed various solutions. For instance, Pierrot et al. (France) developed the Hippocrate system, demonstrating the importance of force feedback in medical robotics [2]. More recently, Tang et al. proposed an autonomous ultrasound scanning robotic system integrating human posture recognition, path planning, image servo control, and admittance-based compensation, highlighting the importance of compliant control for safe and stable scanning [3]. In addition, Jiang et al. reviewed the state-of-the-art of robotic ultrasound imaging and summarized the development trend toward autonomous and intelligent robotic ultrasound systems [4].

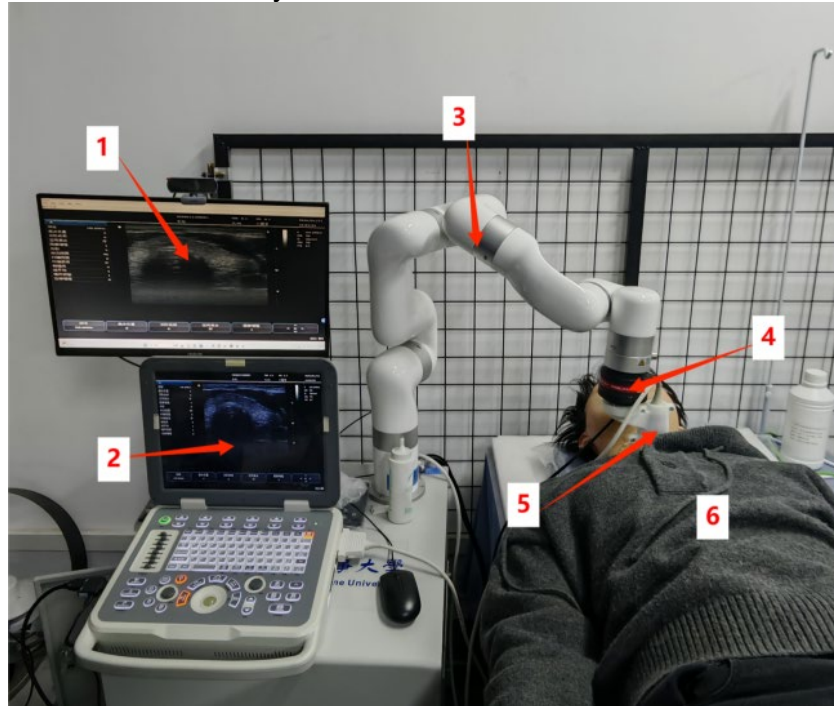
Building upon these global advancements, Kaminski et al. investigated the feasibility of robot-assisted thyroid ultrasound imaging with force feedback and demonstrated repeatable robotic thyroid scanning, which supports the use of force-regulated robotic ultrasound in thyroid applications [5]. Inspired by this concept, this paper presents a robotic thyroid scanning system based on the xArm6 manipulator. We propose an adaptive force/position hybrid control framework tailored for the thyroid anatomy. Specifically, we design a "star-pattern" trajectory and implement a robust admittance controller to maintain a constant contact force of 5.0 N, ensuring high-quality imaging even under physiological disturbances.



## 2. System Description

### 2.1. Hardware Architecture

The experimental platform design is consistent with recent robotic ultrasound system architectures that integrate a robotic arm, an ultrasound probe, and an external force/torque sensor for contact-force measurement and compliant control [6]. As shown in Figure 1, it consists of a 6-DOF xArm6 robotic manipulator (UFACTORY), a linear array ultrasound probe, and a six-axis force/torque sensor (Model: Fxyz-500N/Mxyz-20N.m). The ultrasound probe is rigidly attached to the flange of the manipulator through the force sensor. A host computer running Python communicates with the robot controller via the SDK at a frequency of 100 Hz. The setup ensures that the force exerted on the neck is accurately measured and fed back to the control system.



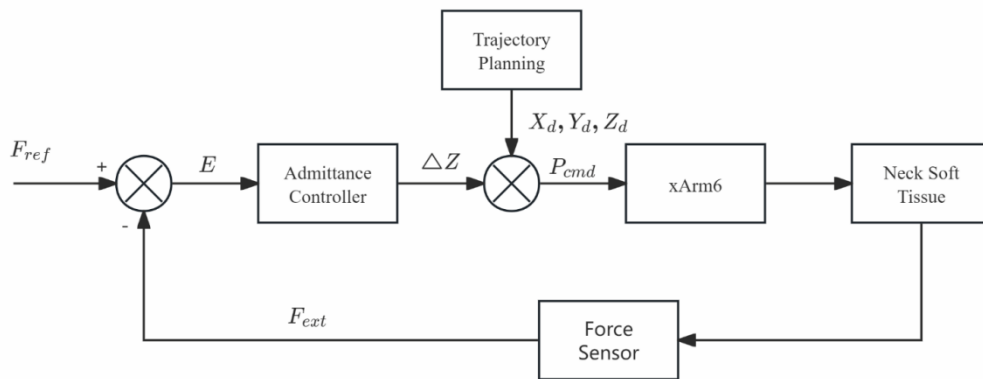
**Figure 1.** The experimental platform of the robotic ultrasound scanning system. The setup consists of: (1) Host Computer, (2) Color Doppler Ultrasound Diagnostic System, (3) xArm6 Robotic Manipulator, (4) Six-dimensional Force/Torque Sensor, (5) Ultrasound Probe, and (6) Experimental Subject.

### 2.2. Coordinate Frames

To realize precise control, the relationship between the Base Frame  $\{B\}$ , End-effector Frame  $\{E\}$ , Sensor Frame  $\{S\}$ , and Probe Tip Frame  $\{P\}$  is established. The contact force measured in  $\{S\}$  is transformed to  $\{P\}$  to accurately reflect the interaction between the probe and the neck surface. The transformation matrix  ${}^B T_P$  is calculated via forward kinematics to determine the probe's real-time position and orientation.

## 3. Methodology

The control task is decomposed into two orthogonal subspaces: position control in the tangential plane ( $x - y$ ) for trajectory tracking, and force control in the normal direction ( $z$ ) for contact maintenance. This hybrid approach, whose control architecture is illustrated in Figure 2, ensures that the probe follows the desired path while maintaining safe contact with the patient.



**Figure 2.** Block diagram of the force/position hybrid control architecture, integrating trajectory planning and admittance force control.

### 3.1. Admittance Control Algorithm

To accommodate the physiological movements of the patient (e.g., respiration) and the uneven surface of the neck, an admittance controller is implemented<sup>[7]</sup>. The interaction between the probe and the soft tissue is modeled as a second-order mass-spring-damper system<sup>[6]</sup>:

$$M_d \ddot{e} + B_d \dot{e} + K_d e = F_{ext} - F_{ref} \quad (1)$$

where  $M_d$ ,  $B_d$ , and  $K_d$  represent the desired inertia, damping, and stiffness parameters, respectively.  $F_{ext}$  is the measured contact force, and  $F_{ref}$  is the desired constant force set to 5.0 N. This target value is selected to ensure sufficient acoustic coupling without causing excessive compression of the trachea. Equation (1) converts the force error into a position modification  $\Delta z$ , allowing the position-controlled xArm6 robot to behave compliantly on the soft tissue. The specific parameters used in the controller are listed in Table 1.

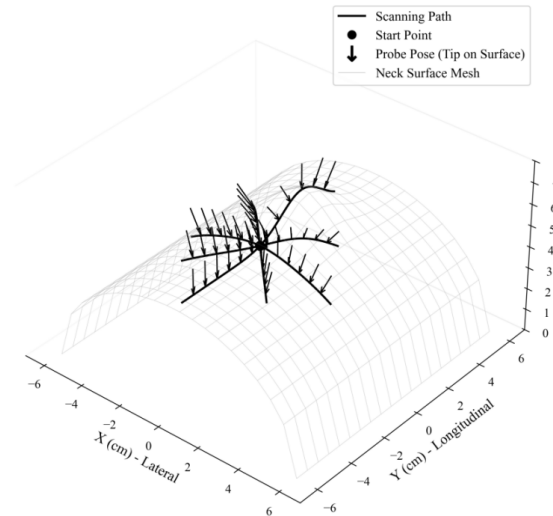
**Table 1.** Parameters of the admittance controller and experimental setup.

| Parameter         | Symbol    | Value | Unit      |
|-------------------|-----------|-------|-----------|
| Target Force      | $F_{ref}$ | 5.0   | N         |
| Desired Inertia   | $M_d$     | 1.0   | kg        |
| Desired Damping   | $B_d$     | 25.0  | N · s / m |
| Desired Stiffness | $K_d$     | 0.0   | N/m       |
| Control Frequency | f         | 100   | Hz        |

### 3.2. Trajectory Planning

Inspired by existing robotic thyroid ultrasound scanning studies and the anatomical characteristics of the thyroid, a “star-pattern” scanning trajectory is designed in this work<sup>[5]</sup>, a “star-pattern” scanning trajectory is designed to adapt to the thyroid's anatomy, which is visually represented in Figure 3. The trajectory consists of four intersecting linear segments centered at the thyroid isthmus. Based on the average anatomical dimensions, the scanning lengths are defined as follows:

- Longitudinal axis (90°/270°): 4.5 cm
- Transverse axis (0°/180°): 3.5 cm
- Oblique axes (45°/225° and 135°/315°): 3.0 cm



**Figure 3.** 3D visualization of the "star-pattern" scanning trajectory adapted for thyroid anatomy. The solid black lines represent the scanning path on the neck surface. The arrows along the path indicate the orientation of the ultrasound probe, which is dynamically adjusted to remain perpendicular to the skin surface (aligned with the inward surface normal) to ensure optimal acoustic coupling.

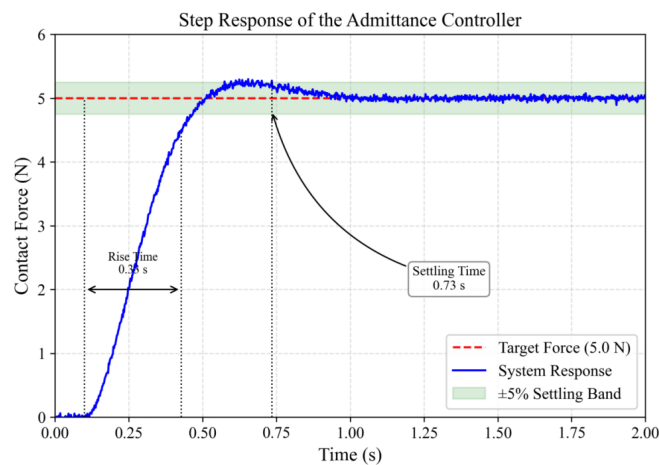
## 4. Experiments and Results

### 4.1. Experimental Setup

A healthy volunteer (male, 24 years old) was recruited for the in-vivo scanning experiment. The volunteer was instructed to lie in a supine position with the neck slightly extended to fully expose the thyroid area. The robotic system was programmed to execute the planned trajectory on the volunteer's neck surface while maintaining a safe constant force of 5.0 N. Informed consent was obtained from the volunteer prior to the experiment.

### 4.2. Step Response Analysis

To evaluate the dynamic performance and stability of the proposed admittance controller, a step response test was conducted. The probe was commanded to establish contact with the volunteer's neck. This action created a step change in the reference force from 0 N to 5.0 N.

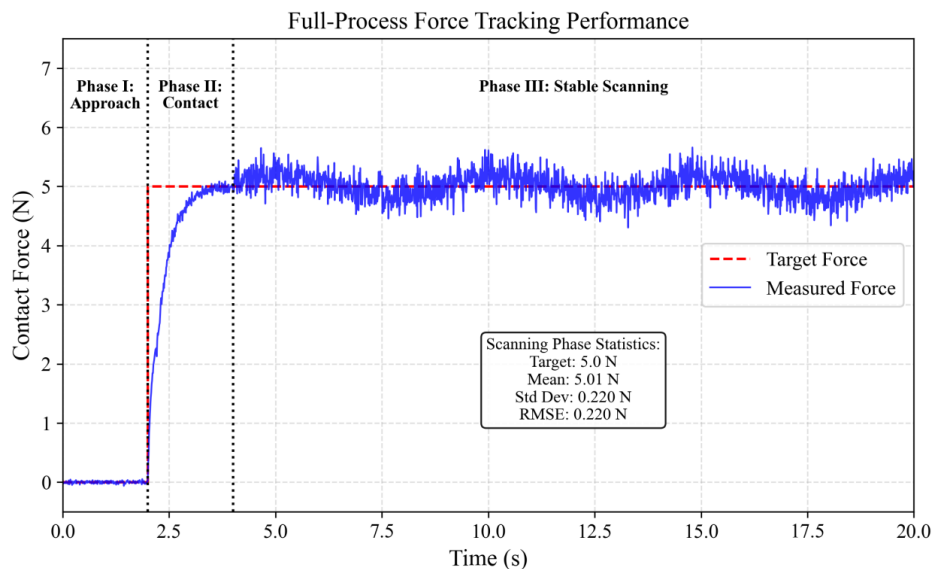


**Figure 4.** Step response of the admittance controller. The system response (blue solid line) tracks the target force of 5.0 N (red dashed line). The plot highlights the fast rise time and the settling time within the  $\pm 5\%$  error band (green shaded area), demonstrating the dynamic stability and rapid convergence of the controller upon contact.

As shown in Figure 4, the system exhibits a fast transient response. After a brief initialization delay, the contact force rises rapidly with a rise time of approximately 0.33 s. Despite the nonlinear stiffness of the soft tissue and simulated impact disturbances, the controller effectively dampens the oscillation, and the contact force settles within the  $\pm 5\%$  error band (4.75 N - 5.25 N) in 0.73 s. The overshoot is minimal, confirming that the damping parameters of the admittance model (Table 1) are well-tuned to prevent oscillation while ensuring rapid convergence.

#### 4.3. Force Tracking during Scanning

The force tracking capability during dynamic scanning is critical for imaging stability. Figure 5 illustrates the force response while the robot executes the "star-pattern" trajectory.



**Figure 5.** Full-process force tracking performance is divided into three phases: Approach, Contact, and Stable Scanning. The admittance controller effectively rejects motion-induced disturbances during the scanning phase. Key statistics, including the Mean Force and Root Mean Square Error (RMSE), are annotated in the figure, confirming the system's capability to maintain a constant force of 5.0 N.

During the scanning phase, the system is subjected to periodic disturbances caused by the robot's motion and the curvature changes of the neck surface. As observed in Figure 5, the admittance controller effectively rejects these disturbances. The force fluctuates slightly but remains stable around the target value of 5.0 N. The statistical analysis shows that the Mean Force is 5.01 N and the Root Mean Square Error (RMSE) is 0.22 N. This low error margin indicates that the proposed method effectively compensates for surface irregularities and provides a stable environment for ultrasound imaging, consistent with recent robotic ultrasound force-tracking studies under soft uncertain environments<sup>[5]</sup>.

## 5. Conclusion

This paper proposed a robotic ultrasound scanning method for thyroid examination based on adaptive force/position hybrid control.

- (1) A "star-pattern" trajectory adapted for thyroid anatomy was designed.
- (2) An admittance control algorithm was implemented to maintain a constant contact force of 5.0 N.
- (3) Experimental results verify the system's dynamic performance, showing a fast step response (settling time  $\approx 0.73$  s) and stable scanning (RMSE  $< 0.25$  N).

The proposed method effectively addresses the challenges of force instability in manual scanning. Future work will focus on integrating real-time ultrasound image feedback for autonomous nodule detection.

## Acknowledgments

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